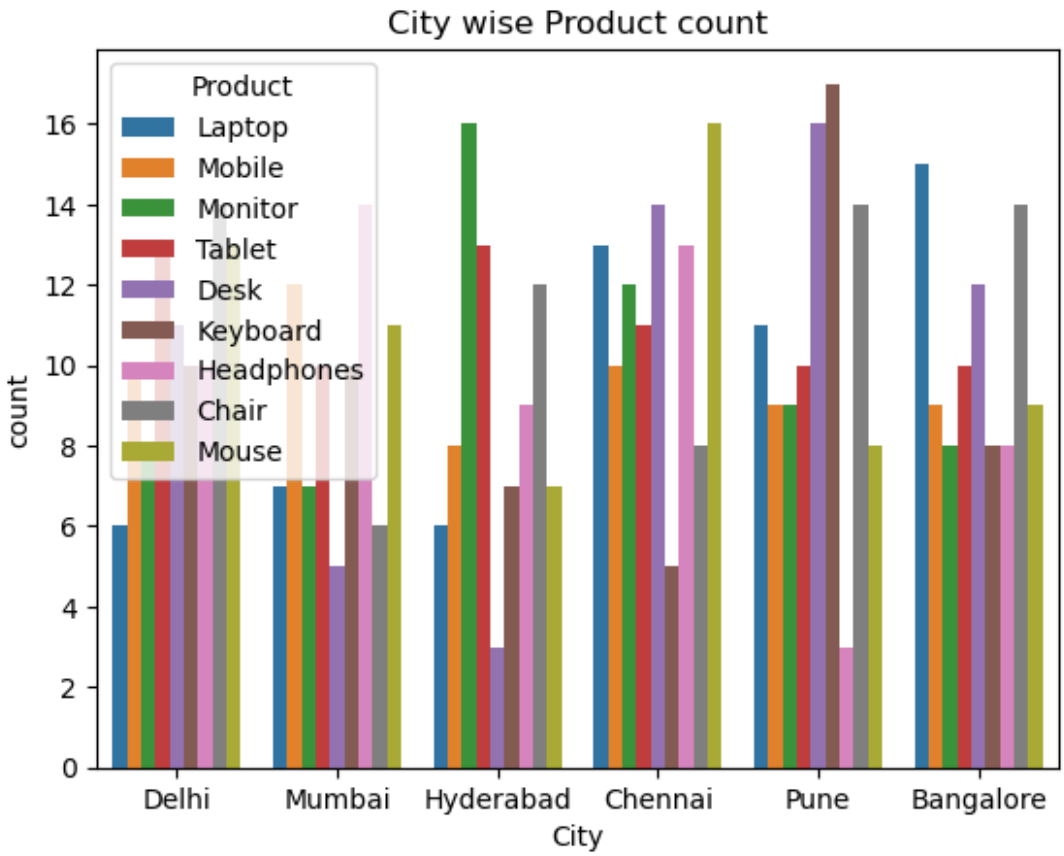
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df=pd.read_csv("Downloads/sales_customer_analytics.csv")

df.dtypes
```

```
OrderID            int64
OrderDate         object
CustomerID         int64
CustomerName      object
City              object
Product           object
Category          object
Quantity           int64
UnitPrice          int64
DiscountPct      float64
PaymentMode       object
OrderStatus       object
Rating           float64
dtype: object
```

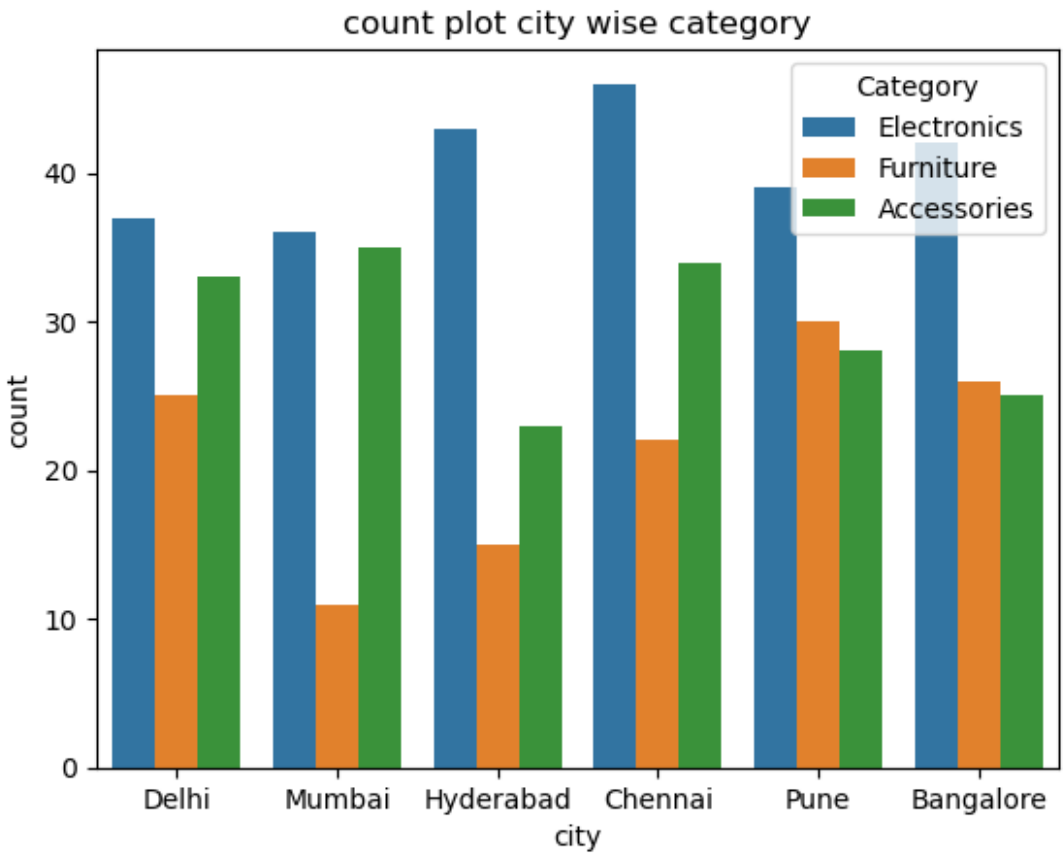
Bivariat

```
sns.countplot(x=df["City"],hue="Product",data=df)
plt.title("City wise Product count")
plt.xlabel("City")
plt.show()
```



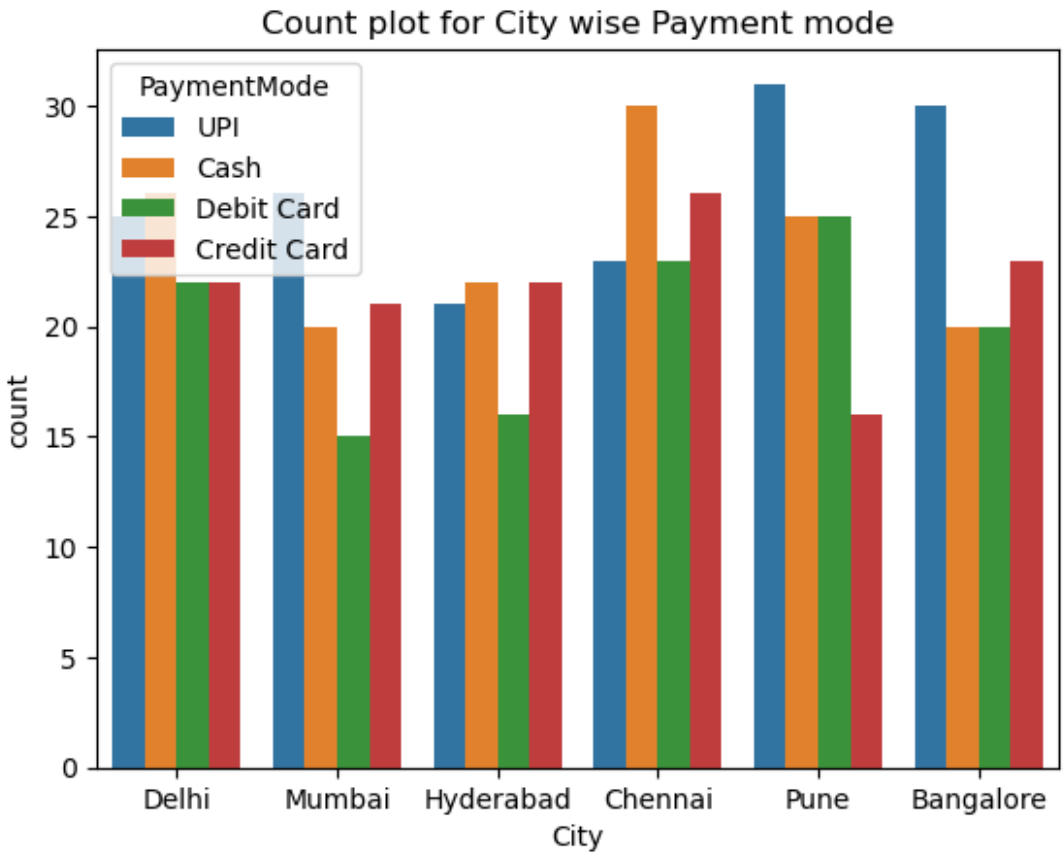
The count plot shows that the highest product keyboard selling in pune city as follow desk in pune,mouse in chennai and monitor in hyderabad and lowest selling product is headphones in pune as follow desk in hyderabad. So, seller should store more stocks of these high selling products as per there city and for low selling products seller can give some discounts on it.

```
sns.countplot(x=df["City"],hue="Category", data=df)
plt.title("count plot city wise category")
plt.xlabel("city")
plt.show()
```



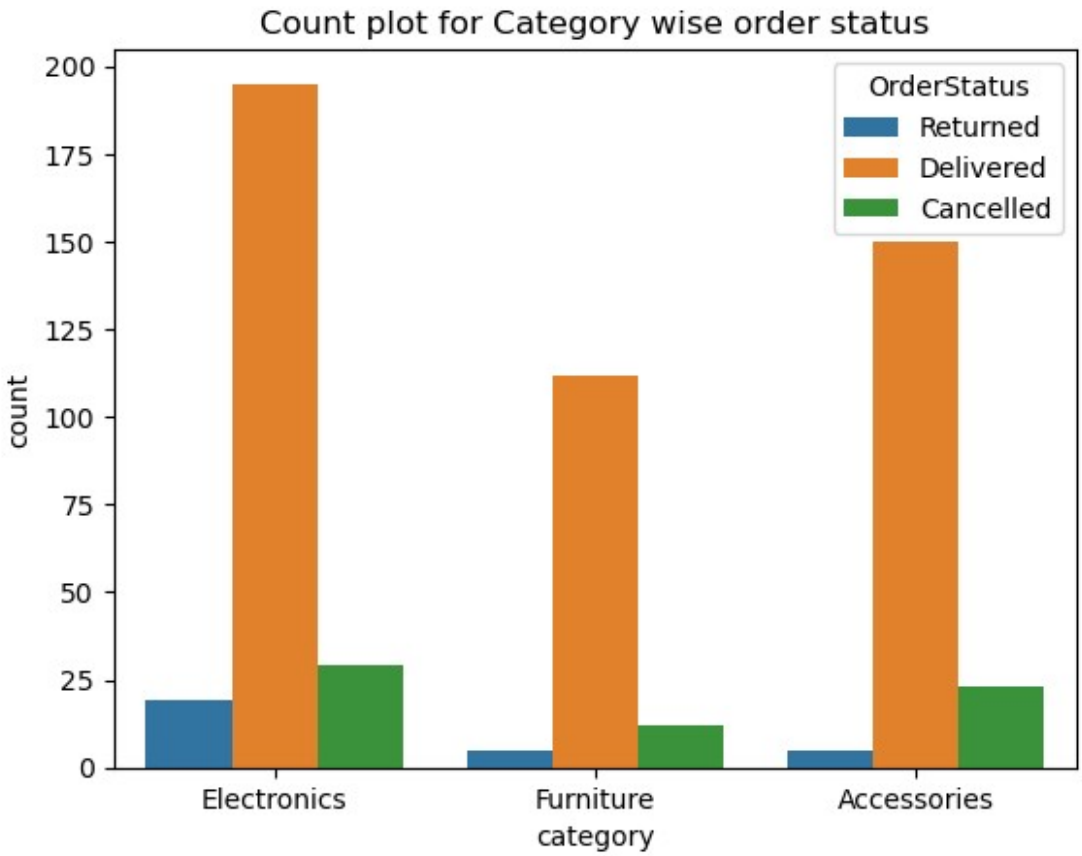
The count plot shows that the highest selling category product is Electronic in Chennai city as follow in hyderabad and lowest category product selling is Furniture in mumbai.So, seller should increase stock of these electronic products in Chennai and Hyderabad and give some discount on furniture products for increasing the sells.

```
sns.countplot(x=df["City"],hue="PaymentMode",data=df)
plt.title("Count plot for City wise Payment mode")
plt.xlabel("City")
plt.show()
```



The count plot shows that the highest payment mode UPI used in pune as followed in Bangalore,cash in chennai and lowest Debit card used in Mumbai and Hyderabad.So, seller can give more discounts as per using of cards during buying products so the user make payments through there's cards.

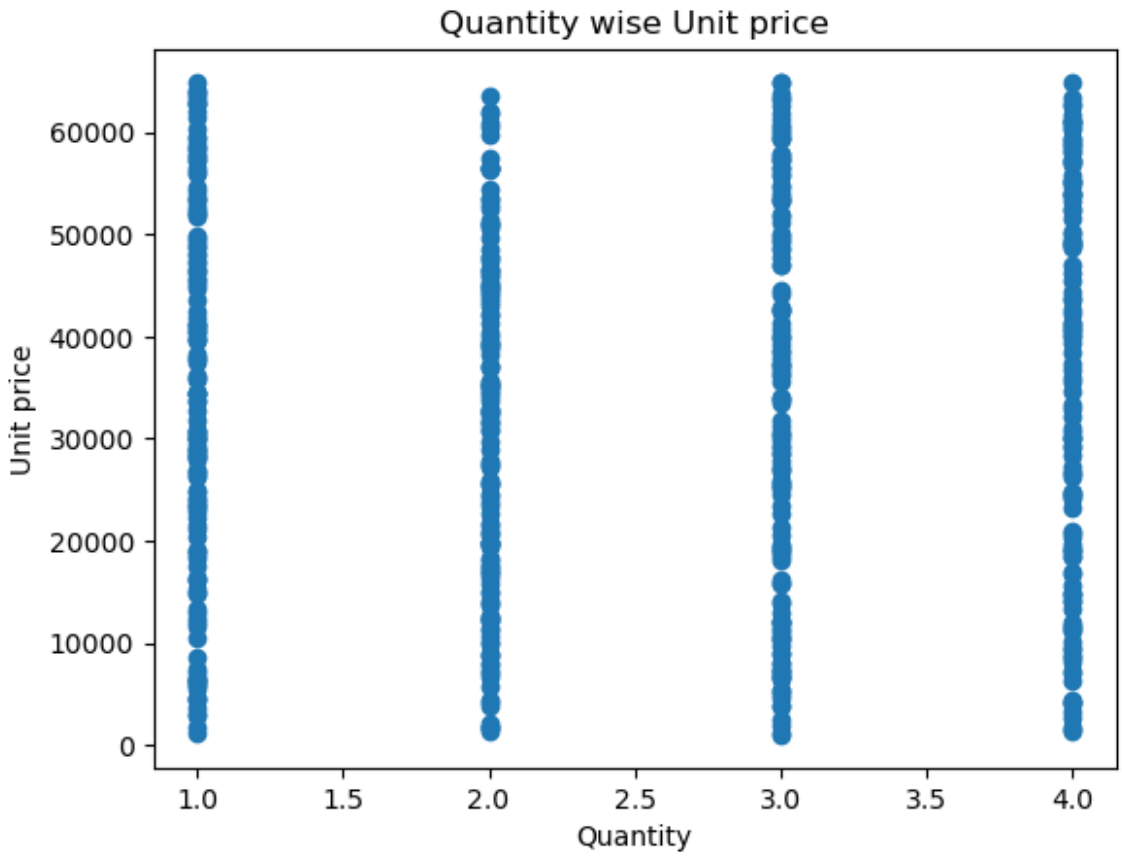
```
sns.countplot(x=df["Category"],hue="OrderStatus",data=df)
plt.title("Count plot for Category wise order status")
plt.xlabel("category")
plt.show()
```



the count plot shows that the highest delivered order is Electronics then accessories , less returned order is Furniture the Accessories and lowest cancelled order is furniture.so, seller should increasing their quality of products and solve customer problems regarding products for increasing furniture sells.

Scatter

```
plt.scatter(df["Quantity"],df["UnitPrice"])
plt.title("Quantity wise Unit price")
plt.xlabel("Quantity")
plt.ylabel("Unit price")
plt.show()
```



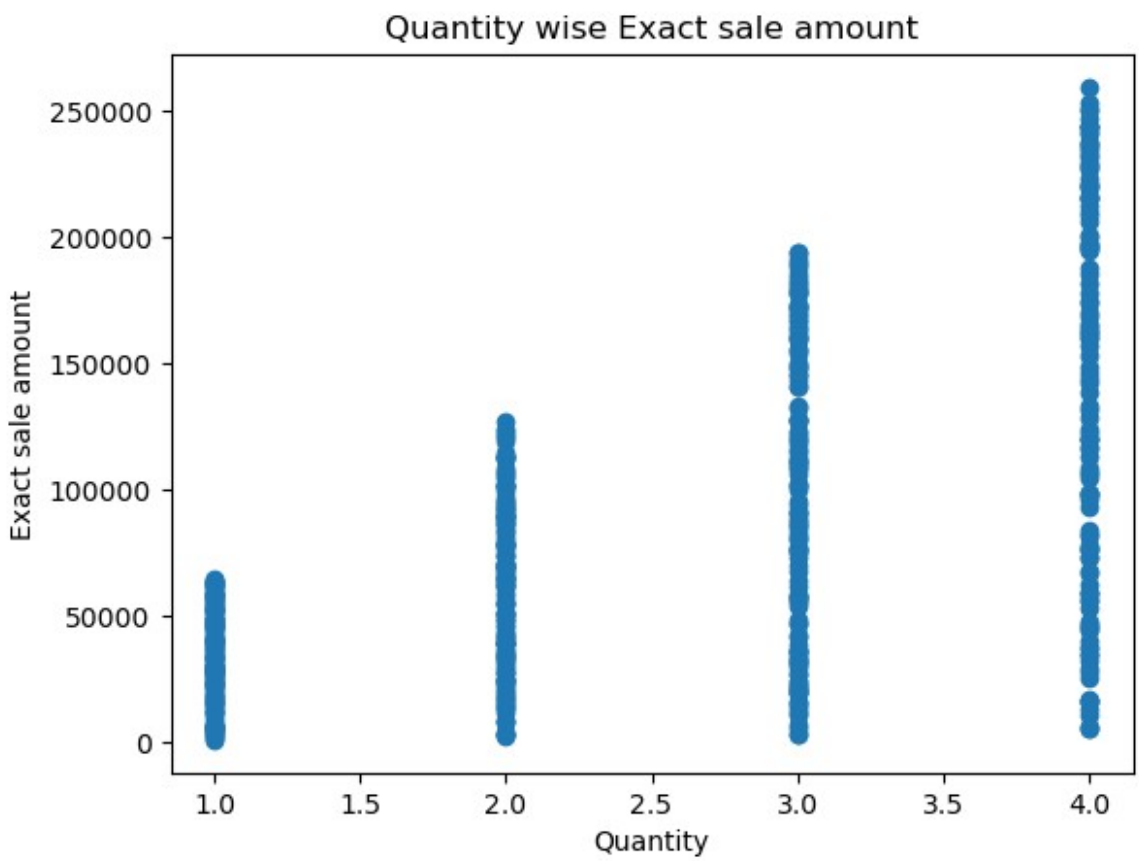
The visualization shows that the relationship between the two features quantity and unit price are uniform trend.

```
df["ExactSaleAmount"]=df["Quantity"]*df["UnitPrice"]

df.head()
```

```
   OrderID   OrderDate  CustomerID CustomerName       City  Product  \
0     1001  2024-01-01         252        Pooja      Delhi   Laptop
1     1002  2024-01-01         293    Siddharth     Mumbai   Mobile
2     1003  2024-01-01         215        Megha  Hyderabad   Mobile
3     1004  2024-01-01         272       Ananya  Hyderabad  Monitor
4     1005  2024-01-02         261        Rohit    Chennai   Tablet

      Category  Quantity  UnitPrice  DiscountPct PaymentMode
OrderStatus  \
0  Electronics         1       5852         0.10         UPI
Returned
1  Electronics         1      27922         0.05        Cash
Delivered
2  Electronics         3      42376         0.15  Debit Card
Cancelled
3  Electronics         1      61492         0.20         UPI
Delivered
```

```
4  Electronics          1       18306          0.15         Cash
Cancelled

   Rating  ExactSaleAmount
0     5.0             5852
1     5.0            27922
2     5.0           127128
3     3.0            61492
4     4.0            18306
```

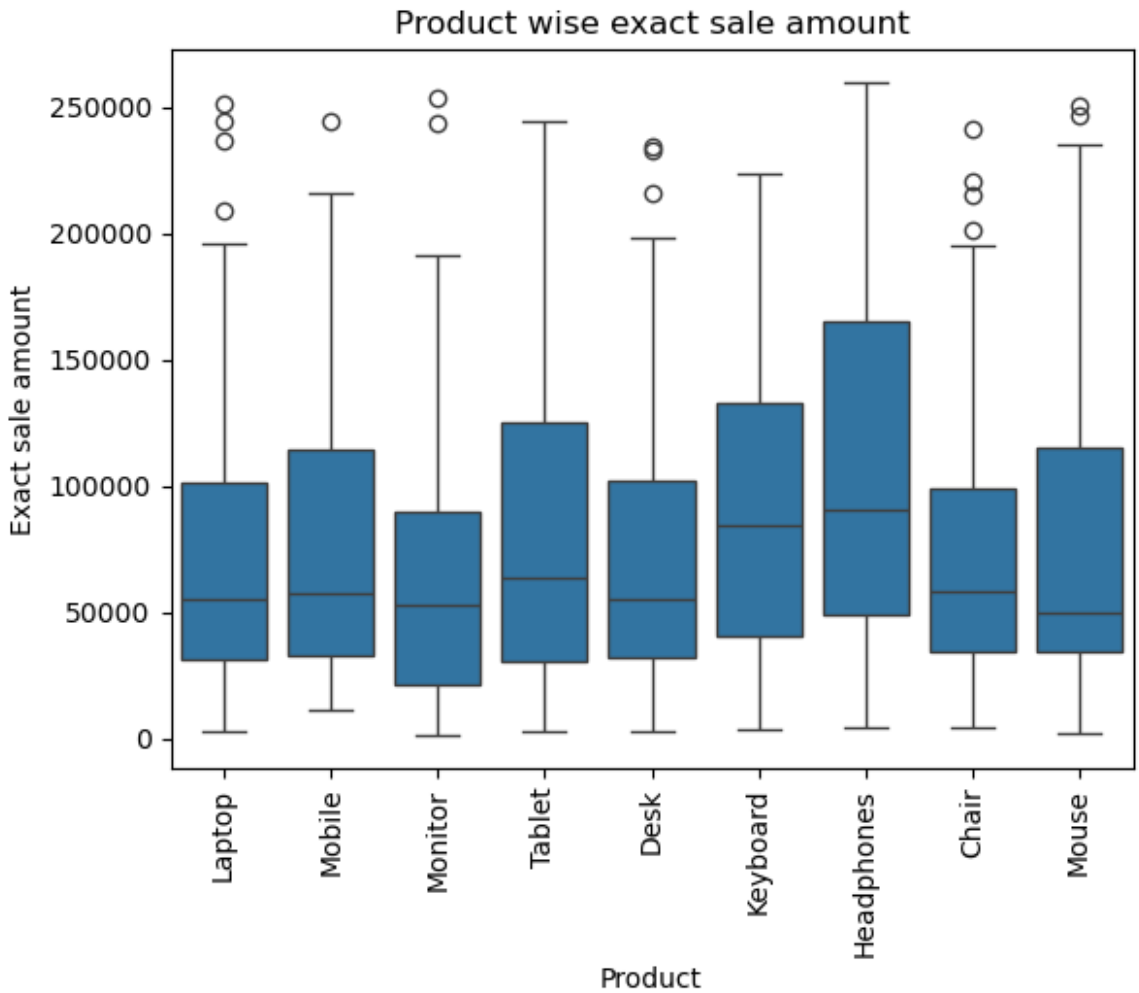
```
plt.scatter(df["Quantity"],df["ExactSaleAmount"])
plt.title("Quantity wise Exact sale amount")
plt.xlabel("Quantity")
plt.ylabel("Exact sale amount")
plt.show()
```



the Visualization shows that the relationship between the two features quantity and ExactSaleAmount is positive as the quantity increases the exact sale amount also increases.There are few orders with 3 and 4 quantity which increases the sale amount for these two quantities.

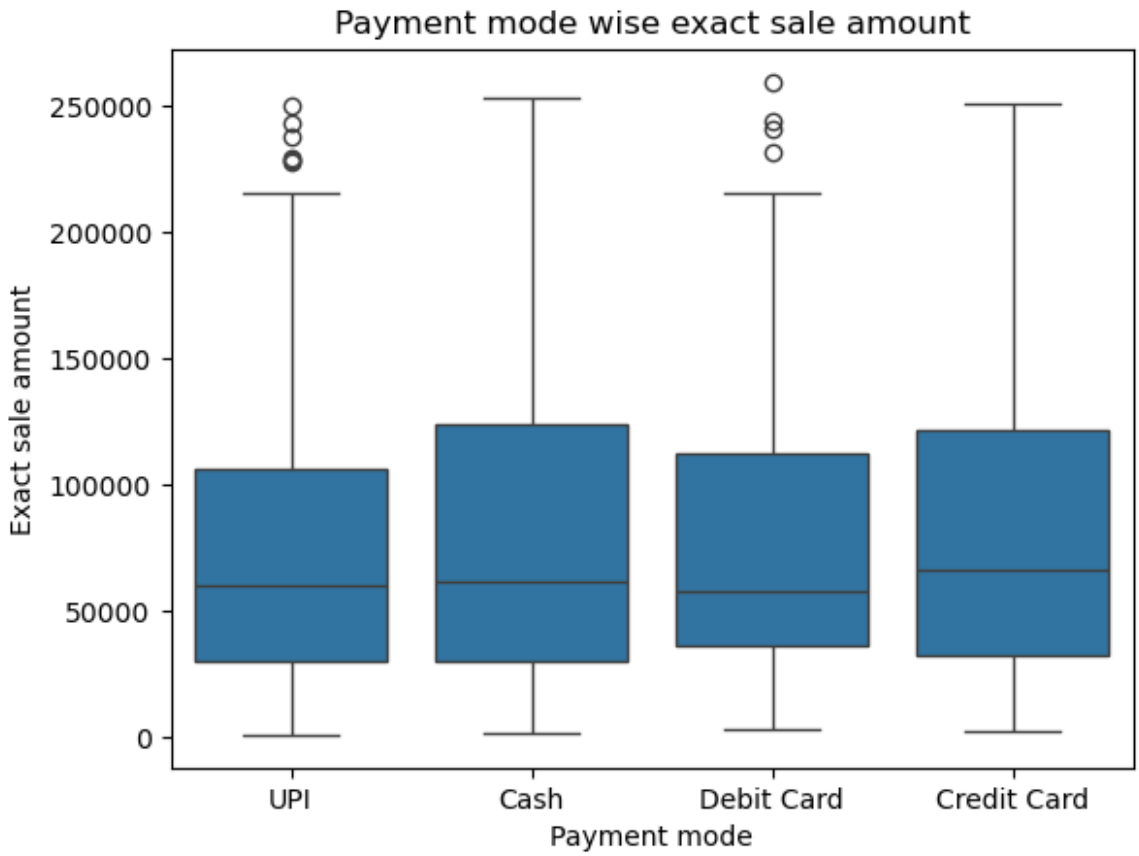
Box plot

```
sns.boxplot(x="Product",y="ExactSaleAmount",data=df)
plt.title("Product wise exact sale amount")
plt.xlabel("Product")
plt.ylabel("Exact sale amount")
plt.xticks(rotation=90)
plt.show()
```



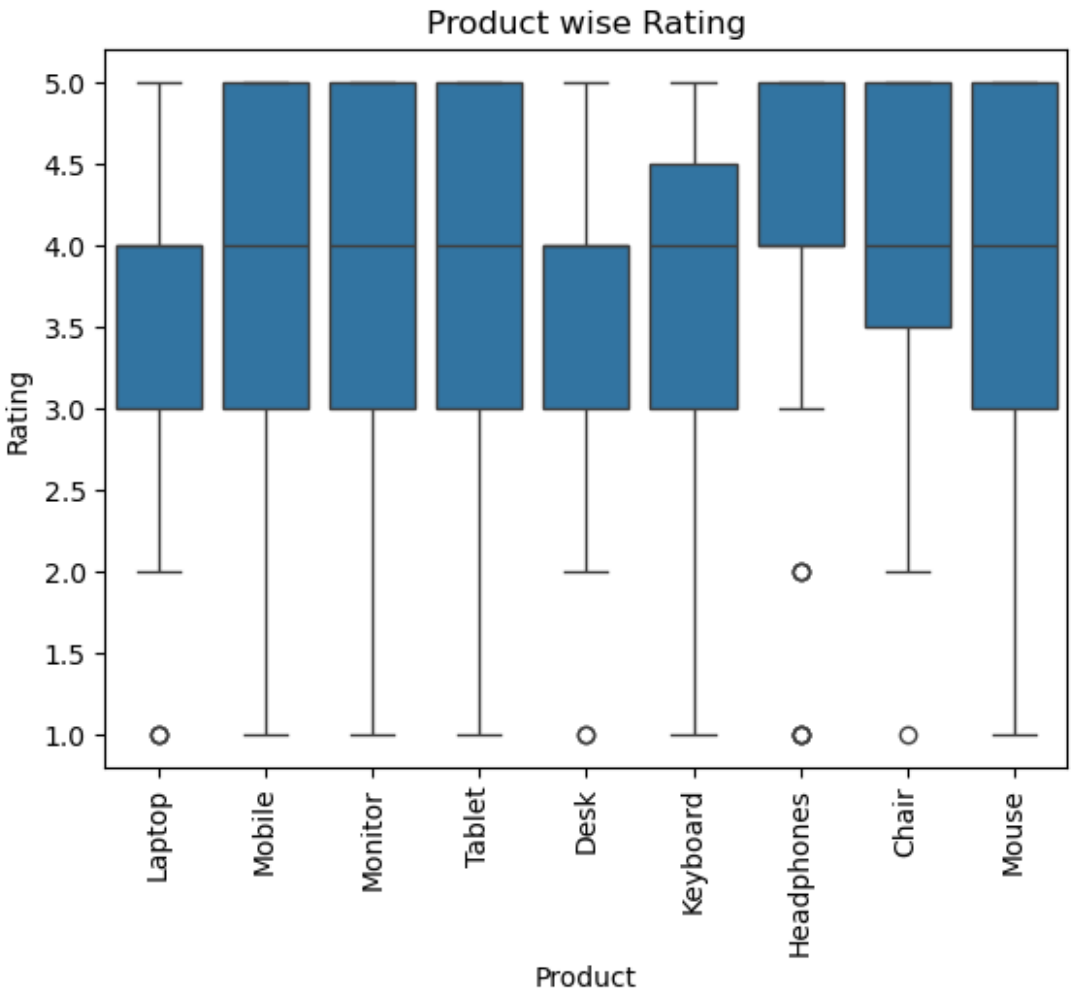
the visualization indicates that the products like headphones,laptops,tablet have higher median sale, whereas the products like mouse, chair,monitor have lower median sales. Also the sales for headphones,keyboard,tablet have no outliers which indicates there are no such orders deriving the sale amount to higher values.

```
sns.boxplot(x="PaymentMode",y="ExactSaleAmount",data=df)
plt.title("Payment mode wise exact sale amount")
plt.xlabel("Payment mode")
plt.ylabel("Exact sale amount")
plt.show()
```



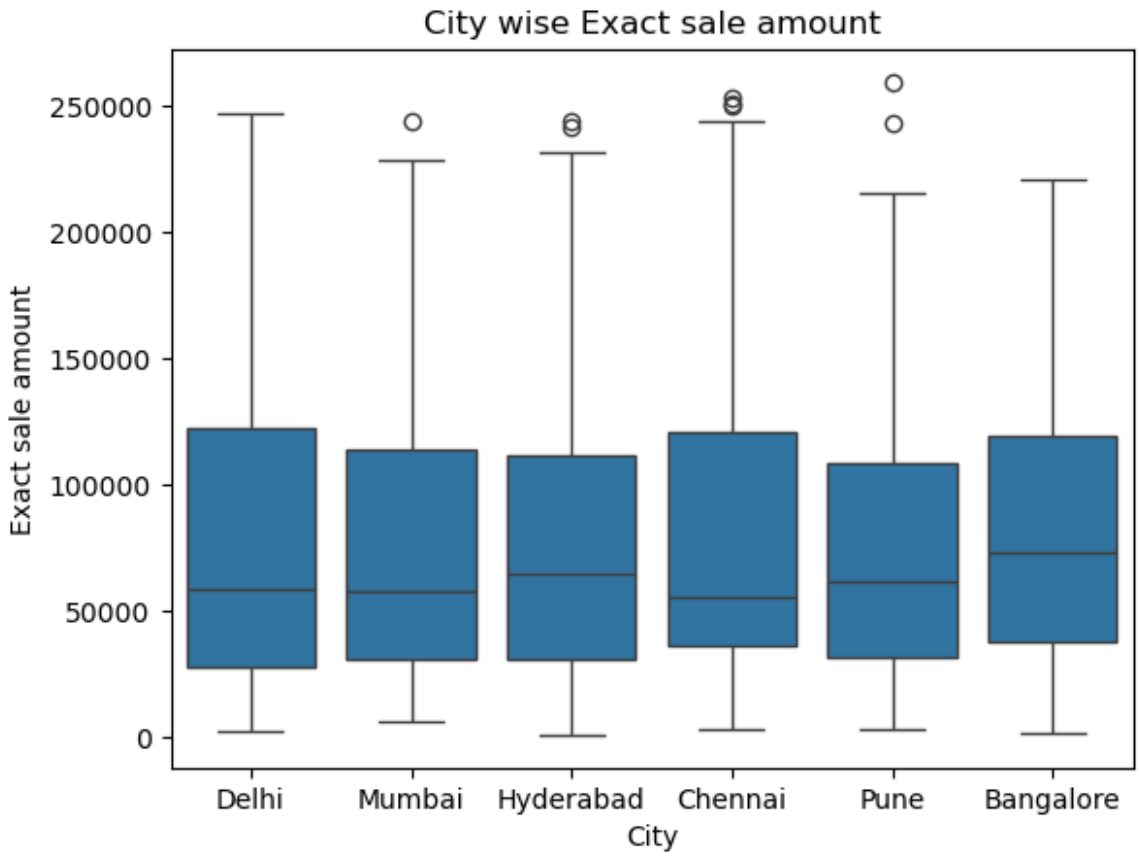
The Visualization indicates that the Payment mode like Credit card have higher median sale and debit card have lower median UPI. Also UPI and Debit card having outliers which indicates there are no such Payment mode deriving the sale amount to higher values.

```
sns.boxplot(x="Product",y="Rating",data=df)
plt.title("Product wise Rating")
plt.ylabel("Rating")
plt.xlabel("Product")
plt.xticks(rotation=90)
plt.show()
```



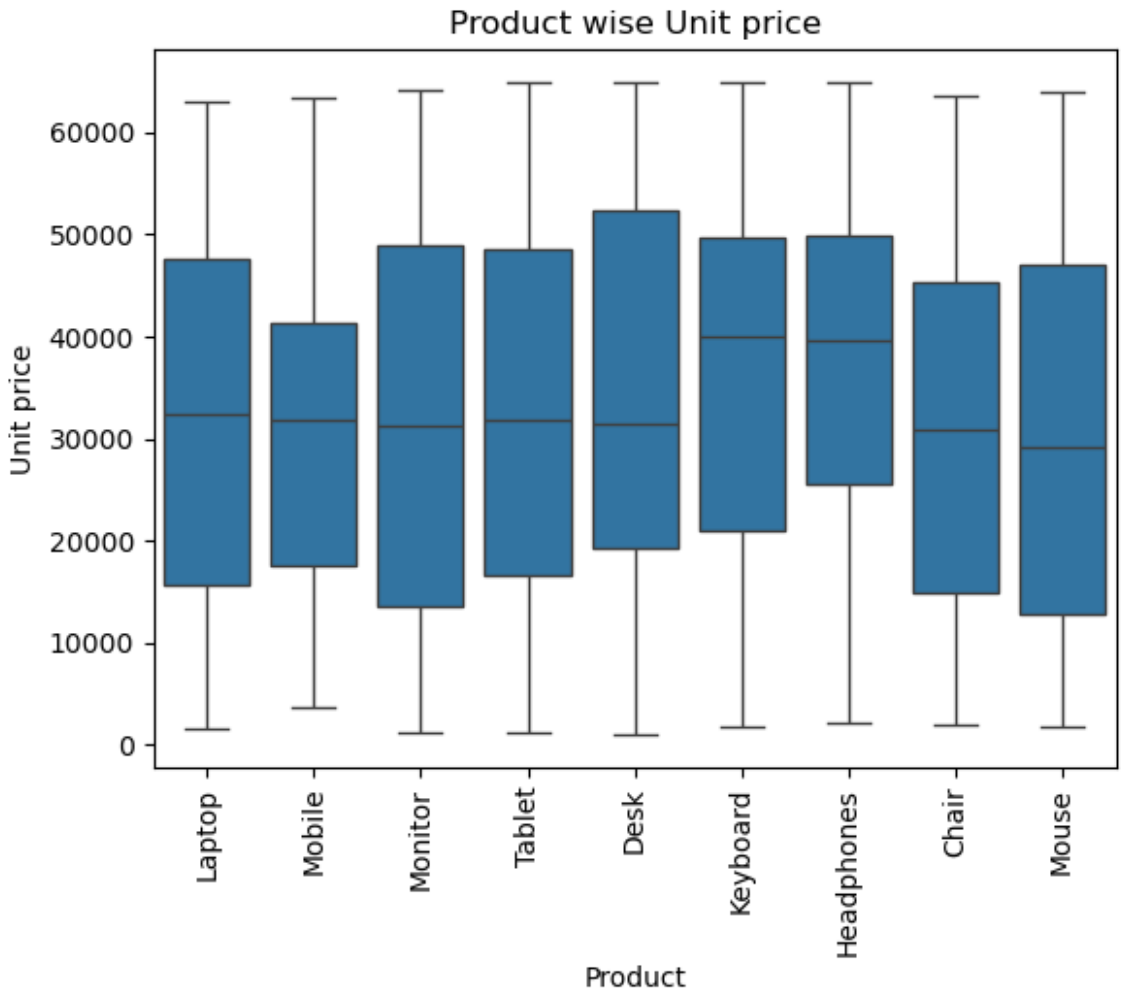
The Visualization indicates that the Rating for products like Mouse,Tablet,Monitor and Mobile have higher median rating and lower median is Laptop,Headphones and desk. Also Laptop,Desk,headphones and Chair having outliers which indicates there are no such Product deriving the sale amount to higher lesser rating.

```
sns.boxplot(x="City",y="ExactSaleAmount",data=df)
plt.title("City wise Exact sale amount")
plt.xlabel("City")
plt.ylabel("Exact sale amount")
plt.show()
```



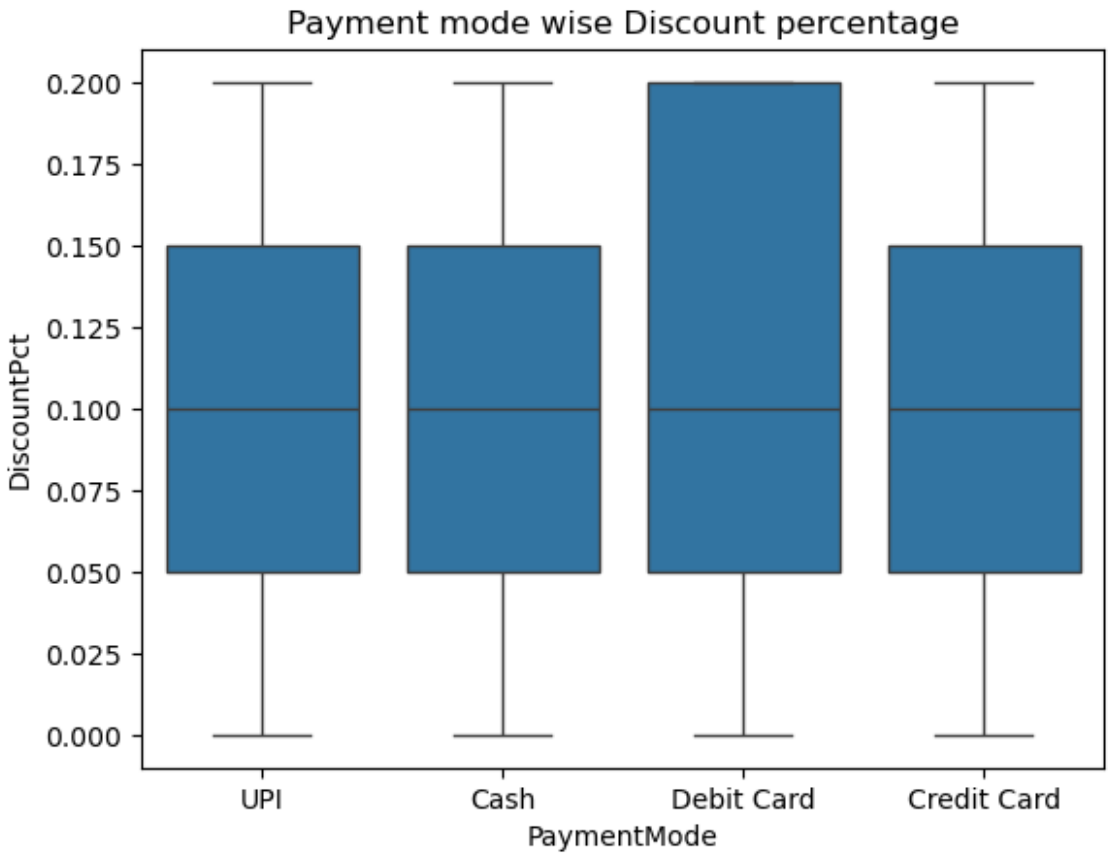
The visualization indicates that higher median city is Pune,Bangalore and hyderabad and lower median city according to exact sale amount is chennai and delhi.

```
sns.boxplot(x="Product",y="UnitPrice",data=df)
plt.title("Product wise Unit price")
plt.xlabel("Product")
plt.xticks(rotation =90)
plt.ylabel("Unit price")
plt.show()
```



The Visualization indicates that the products like Keyboard,headphones and Mobile have higher median rating and lower median is Tablet, mouse and desk.

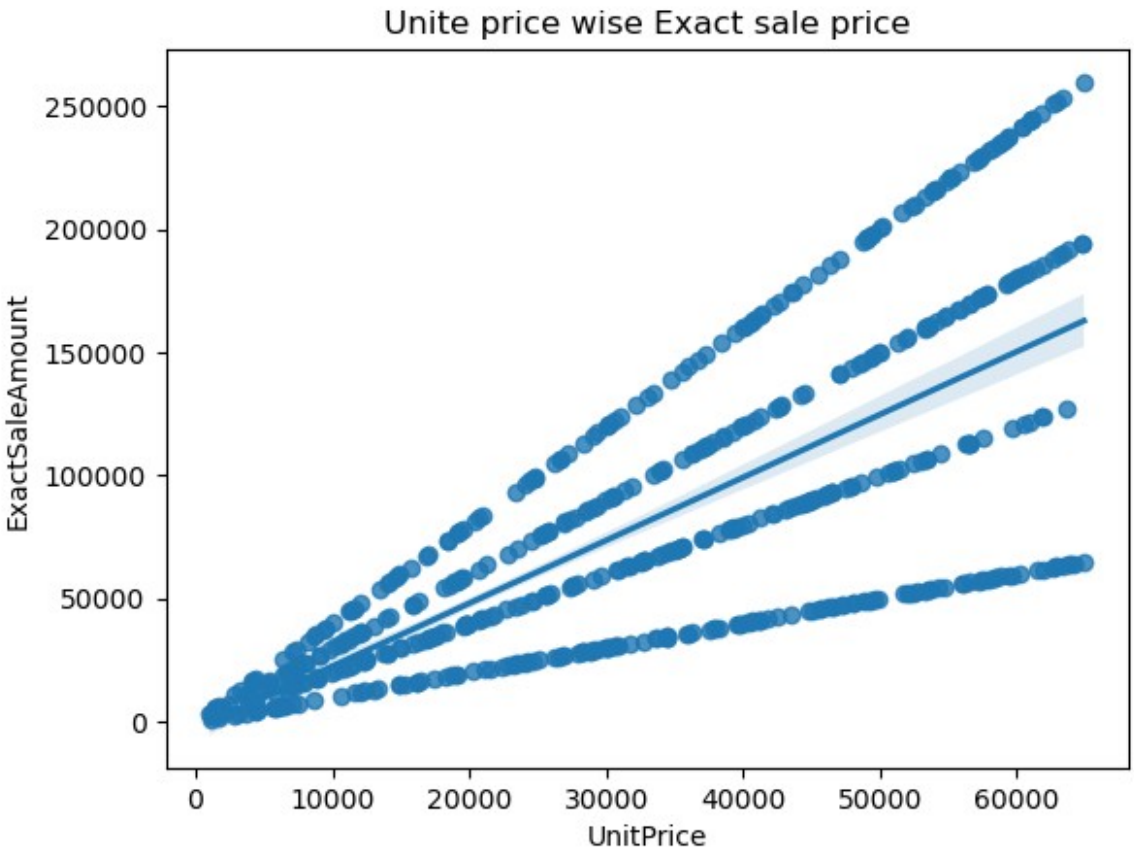
```
sns.boxplot(x="PaymentMode",y="DiscountPct",data=df)
plt.title("Payment mode wise Discount percentage")
plt.show()
```



The Visualization indicates that the paymentmodes like UPI,cash and credit card have higher median rating and lower median is Debit card.

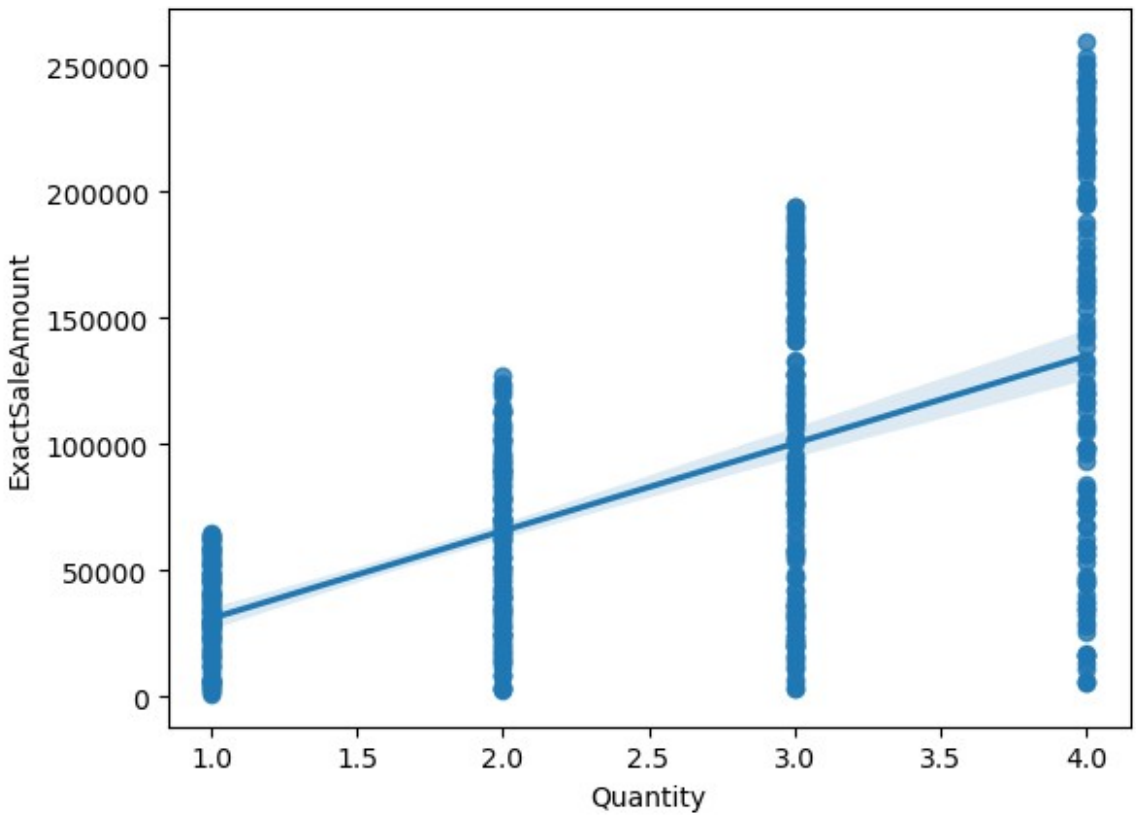
Reggration plot

```
sns.regplot(x="UnitPrice",y="ExactSaleAmount",data=df)
plt.title("Unite price wise Exact sale price")
plt.show()
```



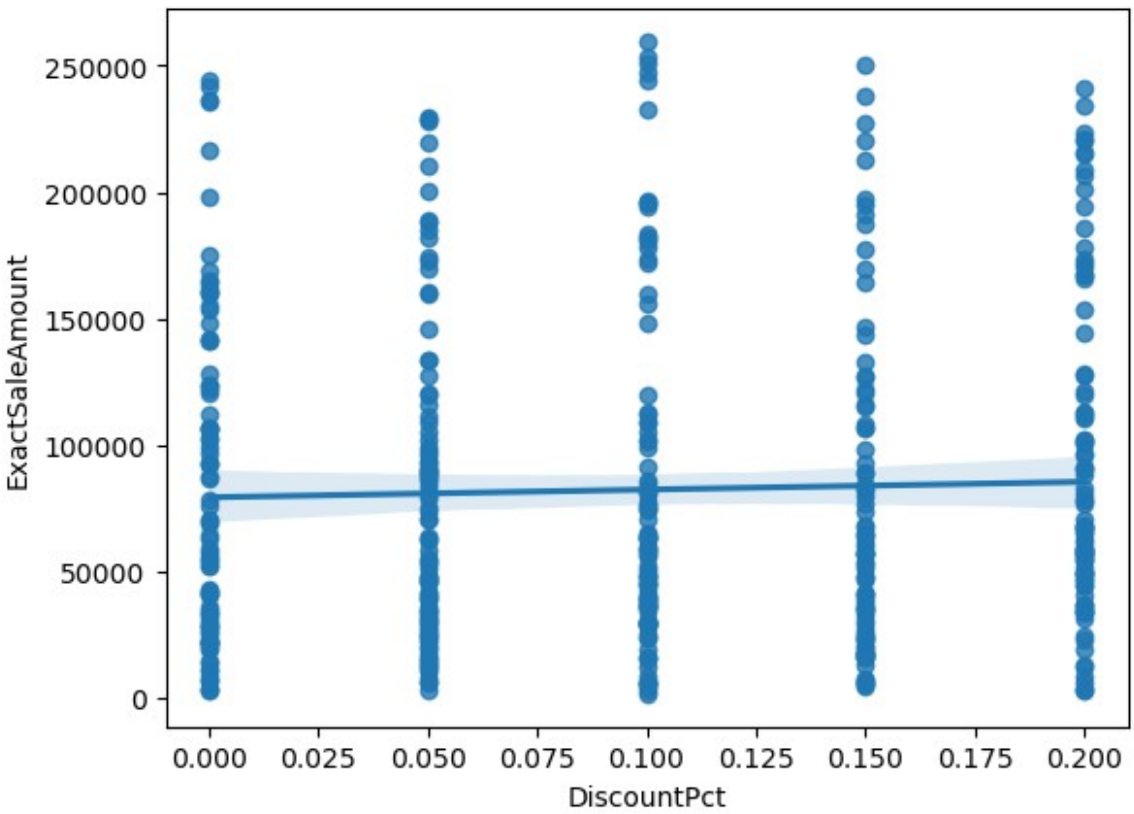
The visualization indicates an upward trend in unit price and exact sale amount feature, which means that there's a positive relationship as the unit price increases the exact sale amount also increases.

```
sns.regplot(x="Quantity",y="ExactSaleAmount",data=df)
plt.title("Quantity wise Exact sale price")
plt.show()
```



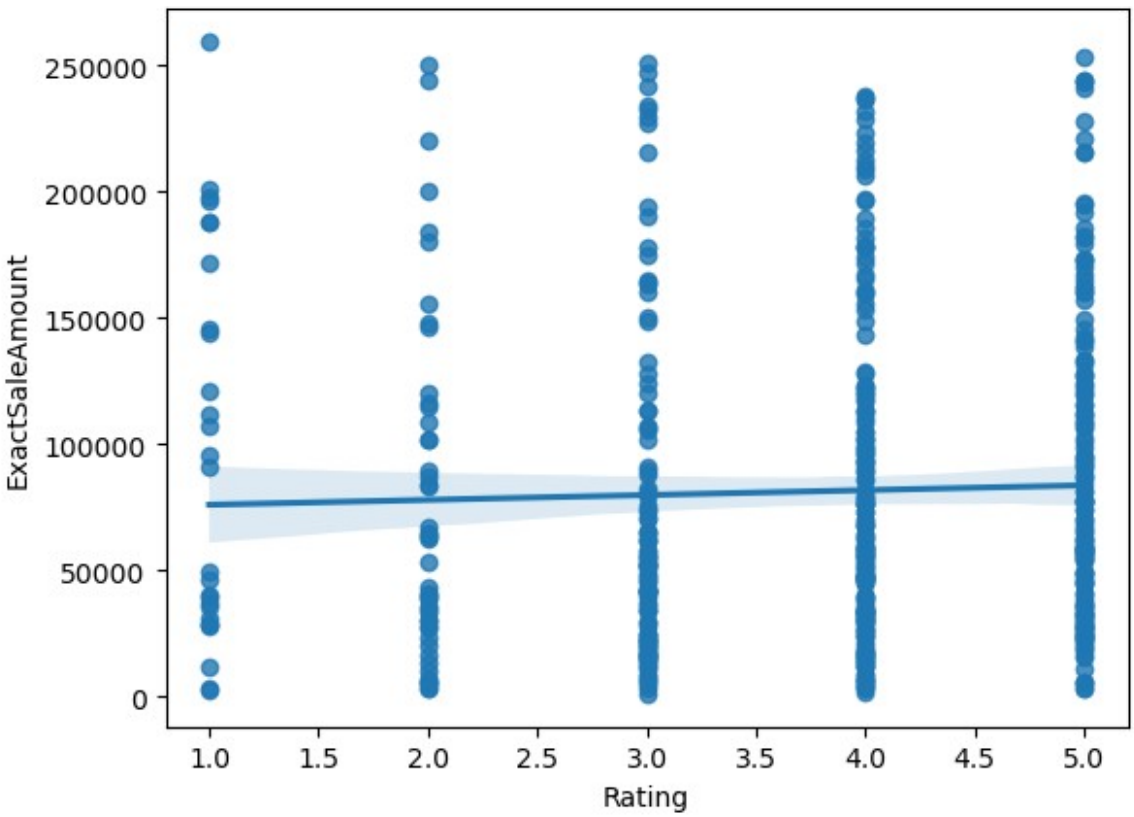
The visualization indicates an upward trend in Quantity and exact sale amount feature, which means that there's a positive relationship as the quantity increases the exact sale amount also increases.

```
sns.regplot(x="DiscountPct",y="ExactSaleAmount",data=df)
plt.title("Discount percentage wise Exact sale price")
plt.show()
```



The visualization indicates an downward trend in discount percentage and exact sale amount feature, which means that there's a negative relationship as the discount percentage decreases the exact sale amount also decreases.

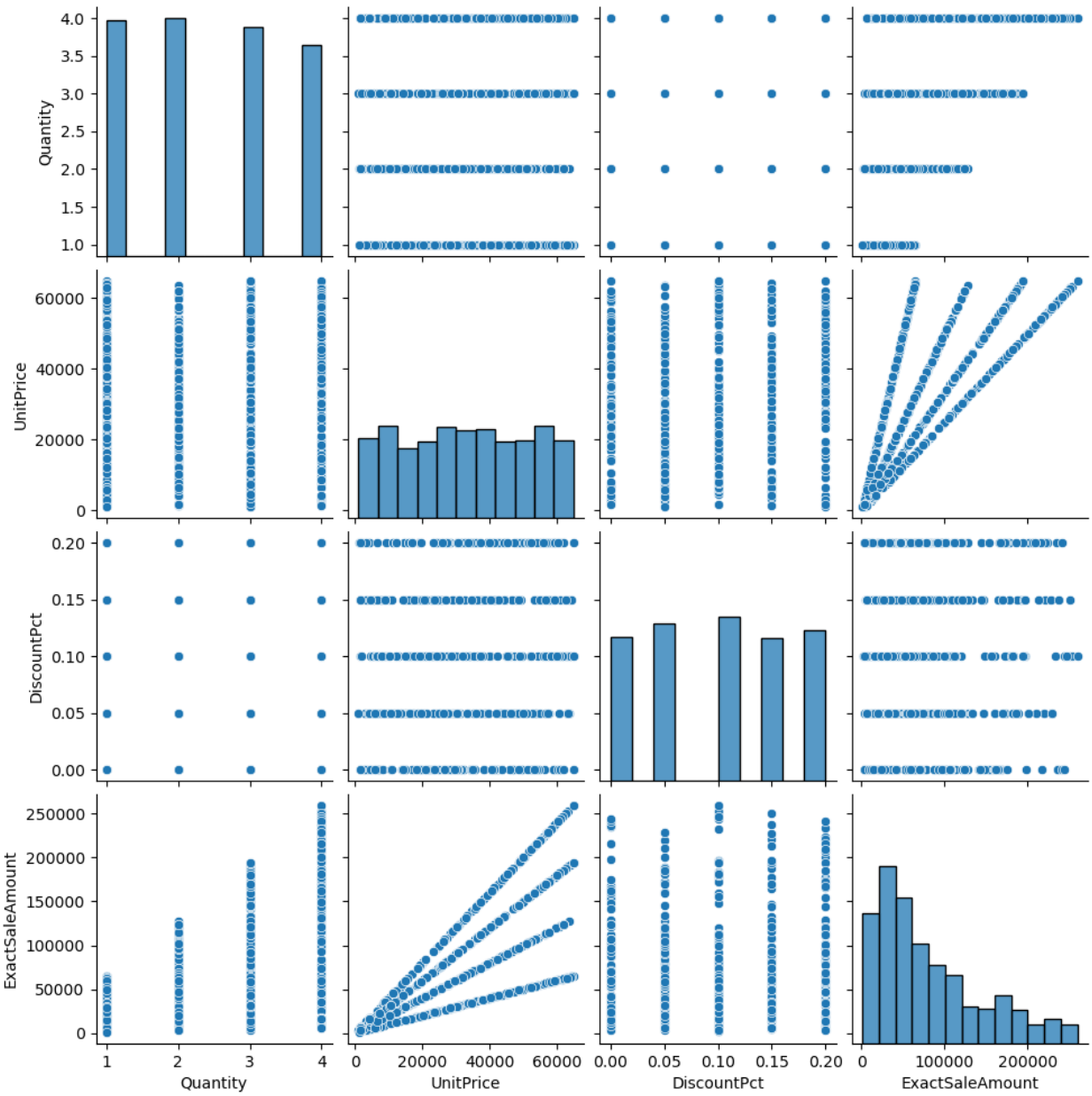
```
sns.regplot(x="Rating",y="ExactSaleAmount",data=df)
plt.title("Rating wise Exact sale price")
plt.show()
```



The visualization indicates an upward trend in rating and exact sale amount feature, which means that there's a positive relationship as the rating increases the exact sale amount also increases.

Pair Plot

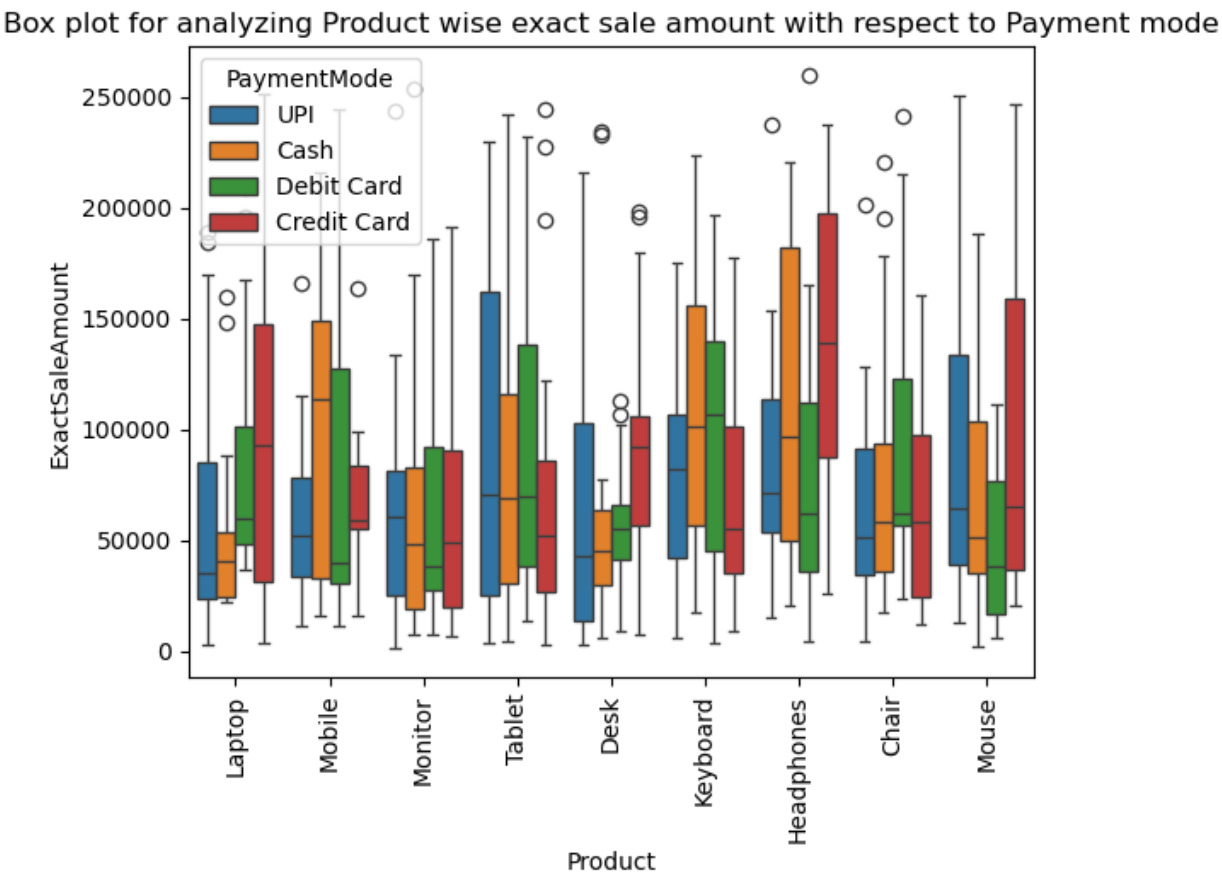
```
sns.pairplot(df[["Quantity","UnitPrice","DiscountPct","ExactSaleAmount
"]])
plt.show()
```



The Visualization shows that the positive trend in between quantity and exact sale amount when quantity increases then exact sale amount also increases and quantity with others features the relationship between then is uniform. -In unit price it shows the positive trend in between unit price and exact sale amount so, when unit price increases then exact sale amount also increases, and unit price with other features the relationship is uniform no changes occur. -In discount percentage there is nothing much changes among these pair graphs. -In exact sale amount it shows positive trend in between exact sale amount and quantity as well positive trend in between exact sale amount and unitprice.

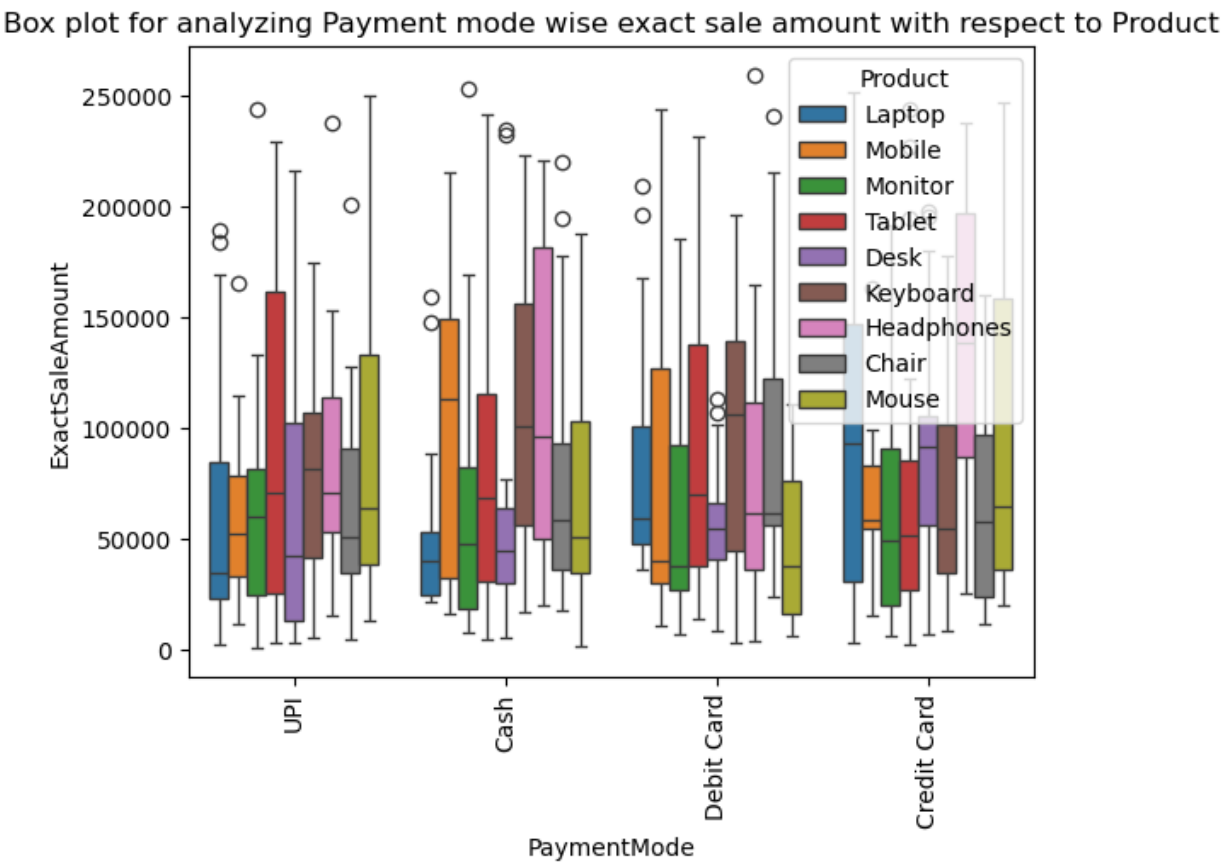
box plot (bivariate)

```
sns.boxplot(x="Product",y="ExactSaleAmount",hue="PaymentMode",data=df)
plt.title("Box plot for analyzing Product wise exact sale amount with
respect to Payment mode")
plt.xticks(rotation=90)
plt.show()
```



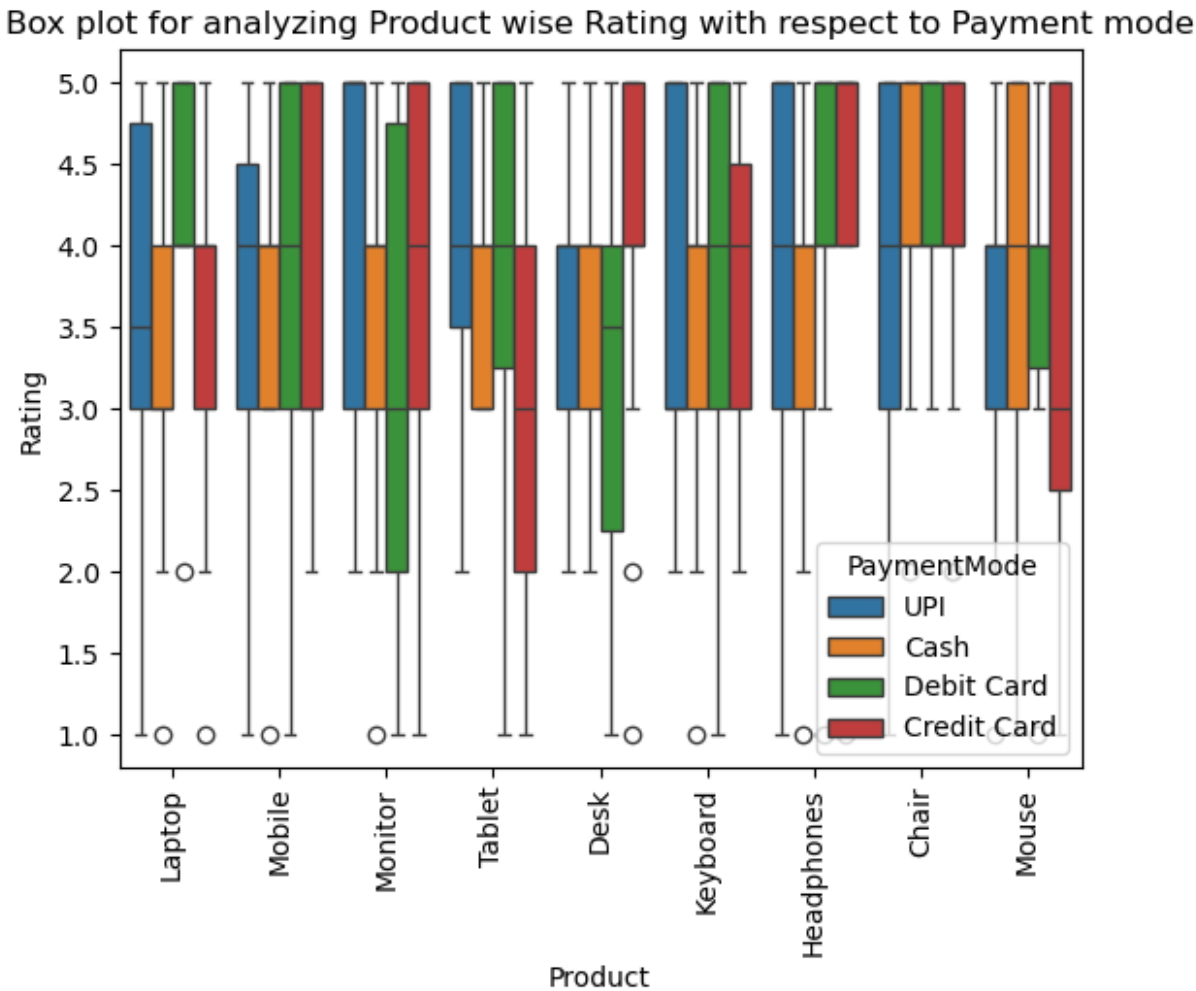
The visualization shows that the highest median payment mode of an product like credit card in headphones and laptop as follow debit card in keyboard and cash in mobile and lowest median are debit card in chair and in mobile and laptop.

```
sns.boxplot(x="PaymentMode",y="ExactSaleAmount",hue="Product",data=df)
plt.title("Box plot for analyzing Payment mode wise exact sale amount
with respect to Product")
plt.xticks(rotation=90)
plt.show()
```



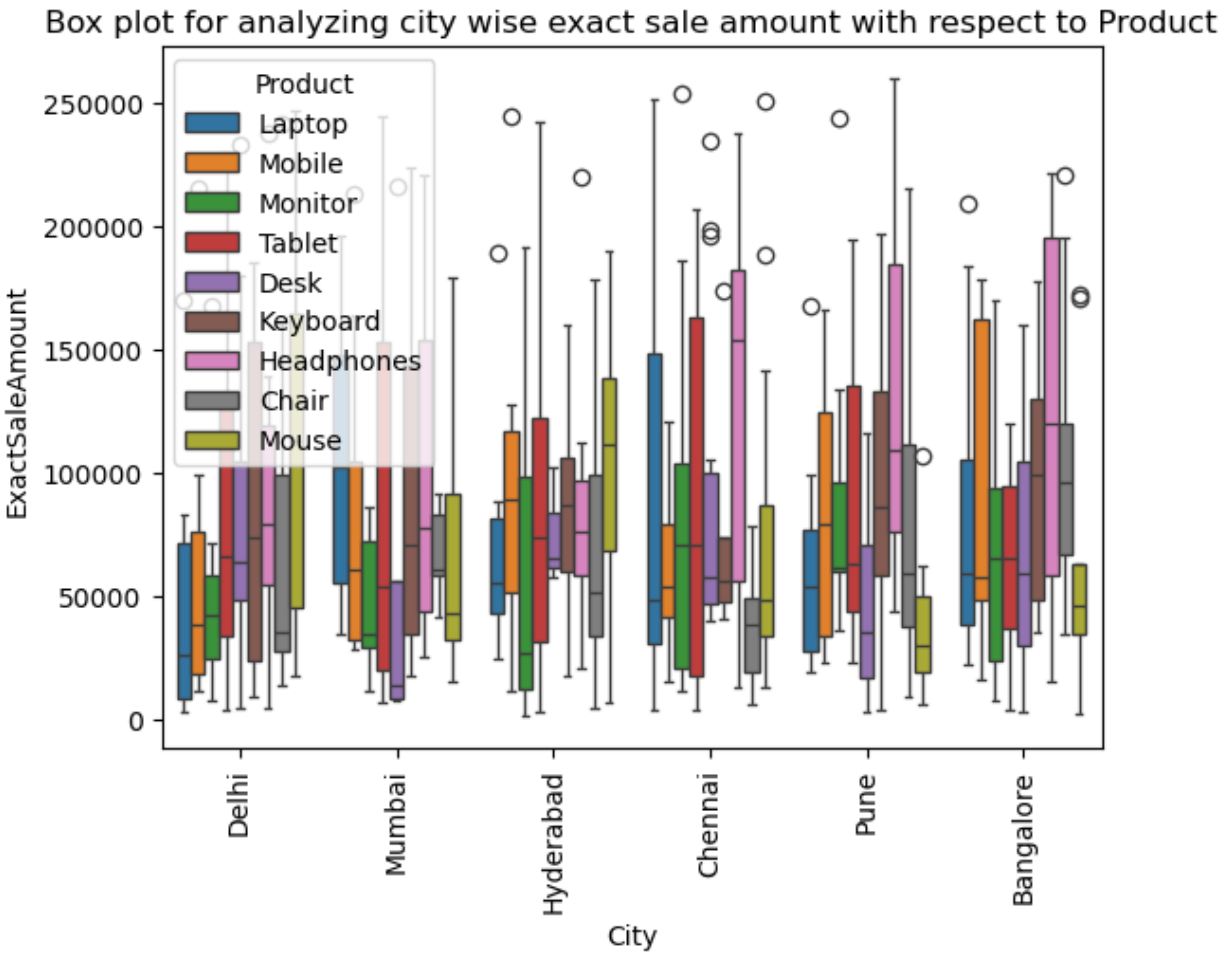
The Visualization shows that the highest median of product as per payment mode like in UPI monitor,keyboard,in cash the product mobile,laptop ,in debit card the product keyboard and headphones and in credit card product laptop and headphones and lowest median payment mode of product is in UPI mobile,chair ,laptop and mouse, in cash product is mouse,desk,and chair , in debit card product is chair,mobile,laptop, and desk and in credit card product is mobile.

```
sns.boxplot(x="Product",y="Rating",hue="PaymentMode",data=df)
plt.title("Box plot for analyzing Product wise Rating with respect to
Payment mode")
plt.xticks(rotation=90)
plt.show()
```



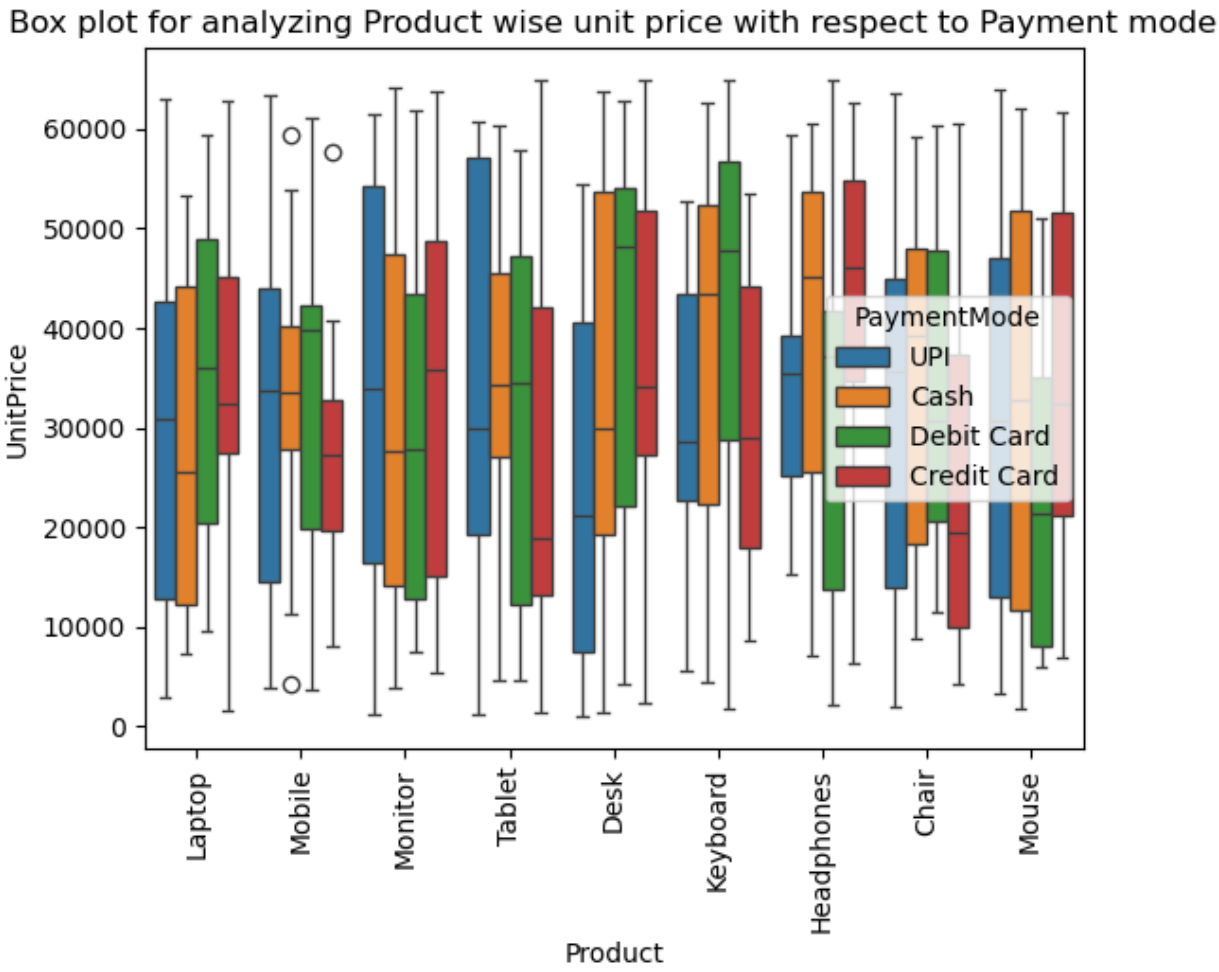
the visualization shows that the highest rating median of product desk in debit card as follow mobile in UPI and lowest median of product laptop of UPI as follow mouse in credit card. also there is some outliers in all products.

```
sns.boxplot(x="City",y="ExactSaleAmount",hue="Product",data=df)
plt.title("Box plot for analyzing city wise exact sale amount with
respect to Product")
plt.xticks(rotation=90)
plt.show()
```



the visualization shows that as per exact sale amount highest median of city as per products like hyderabad in mobile, mouse, in chennai product headphones, in pune product is monitor , in pune products is laptop and bangalore product is monitor and keyboard. also outliers are everywhere for increasing median of product as per exact sale amount.

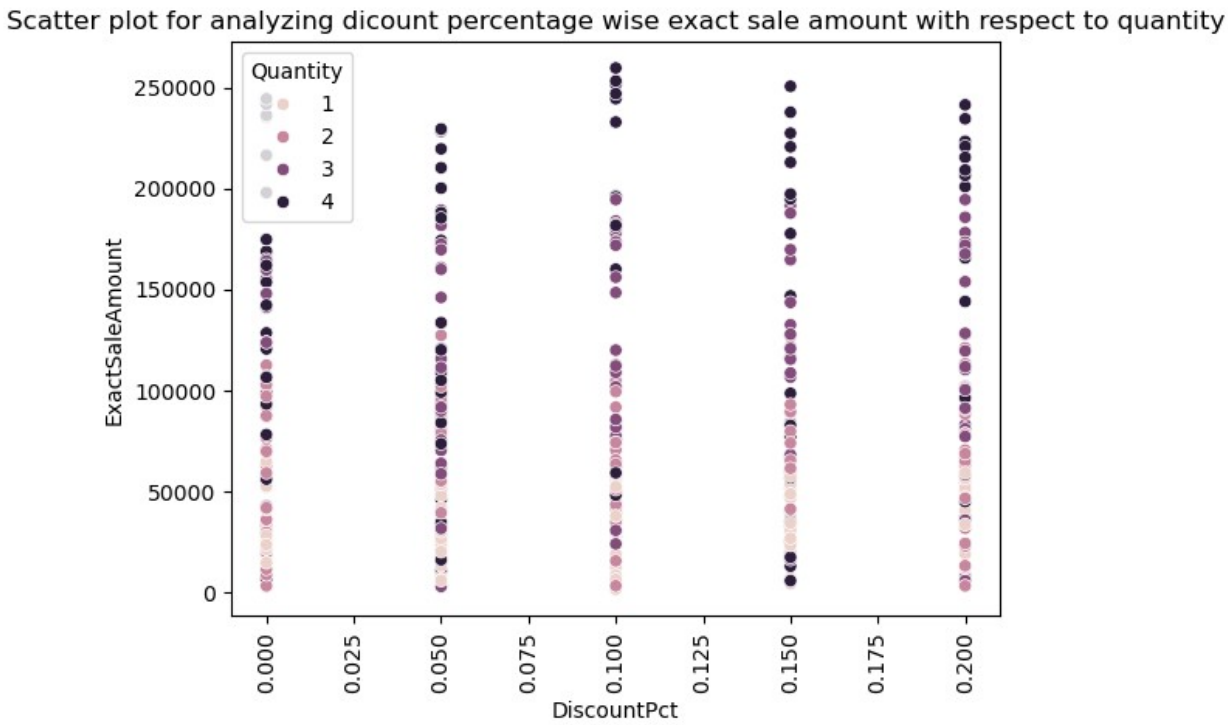
```
sns.boxplot(x="Product",y="UnitPrice",hue="PaymentMode",data=df)
plt.title("Box plot for analyzing Product wise unit price with respect
to Payment mode")
plt.xticks(rotation=90)
plt.show()
```



The visualization shows that the unit price highest median of product wise payment mode like in laptop UPI, debit card , in mobile UPI and debit card, monitor credit card, tablet debit card, desk debit card, keyboard cash and debit card , headphones cash and credit card, chair cash and cash,and lowest median like in laptop-credit card, mobile-cash,monitor-cash,tablet-UPI,cash,credit card , in desk-UPI,credit card, in keyboard-UPI,chair-credit card and mouse debit card and credit card. also some outliers in mobile product in cash and credit card payment mode.

Scatter plot (bivariate)

```
sns.scatterplot(x="DiscountPct",y="ExactSaleAmount",hue="Quantity",dat
a=df)
plt.title("Scatter plot for analyzing dicount percentage wise exact
sale amount with respect to quantity")
plt.xticks(rotation=90)
plt.show()
```



the Visualization indicates there is no certain trend between discount percentage and exact sale amount,but we can say if the discount is higher the sale amount gets lowered compared to other discount percentages.There's not a single quantity dominating the discount,but higher discounts are usually given to higher quantities.lower quantities get discount around 5%-10%.

```
sns.scatterplot(x="UnitPrice",y="ExactSaleAmount",hue="Quantity",data=
df)
plt.title("Scatter plot for analyzing unit price wise exact sale
amount with respect to quantity")
plt.xticks(rotation=90)
plt.show()
```

The Visualization shows that the positive trend between unitprice and exact sale amount features with respect to quantity. It means when quantity increases the exact sale amount also increases.

Amazon video prime DataFrame

```
df=pd.read_csv("Downloads/amazon_prime_video_data.csv")

df.dtypes
```

```
content_id              int64
content_type           object
genre                  object
language               object
release_year            int64
duration_minutes        int64
imdb_rating           float64
age_rating             object
country                object
added_date             object
views_millions        float64
avg_watch_time_min    float64
likes_millions        float64
subscription_plan      object
is_trending            object
```